

FlexBox 1.2.1 Backend Cheatsheet

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2026-02-24

Contents

1 FlexBox 1.2.1 Uplink/Downlink Schema	1
1.1 FPorts (uplink)	1
1.2 Uplink: Event codes and payloads	2
1.2.1 0x00 — Button (public vote)	2
1.2.2 0x01 — NFC check-in	2
1.2.3 0x02 — NFC check-out	2
1.2.4 0x03 — NFC registered vote	3
1.2.5 0x10 — Battery status (heartbeat)	3
1.2.6 0x12 — Counter sync	4
1.2.7 0x11 — Low battery Alarm (reserved)	4
1.2.8 0xFF — Future	4
1.3 Downlink: Commands	4
1.4 Notes	5
1.5 Example payloads (real uplinks)	6

1 FlexBox 1.2.1 Uplink/Downlink Schema

Reference for backend; Payload layout, FPorts, and downlink commands. Device is LoRaWAN Class A, OTAA; max payload **11 bytes**. All uplink timestamps are **UTC epoch seconds, big-endian**, bytes 0–3.

1.1 FPorts (uplink)

FPort	Use
10	Button (public vote)
11	NFC (check-in, check-out, registered vote)
20	Housekeeping (battery, counter-sync)
30	Alarms (reserved)
13	Future

Downlink: Device uses **first byte = command**; FPort is not used for routing. Any FPort is accepted.

1.2 Uplink: Event codes and payloads

Common: [0..3] = timestamp (uint32 BE), [4] = event code. Rest = event-specific. All 11 bytes.

1.2.1 0x00 — Button (public vote)

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x00
button_id	5	0..5
counter	6–8	24-bit BE (per-button)
reserved	9–10	0

- **FPort:** 10
 - **When:** User presses a feedback button.
 - **Example:** 69 38 6F 33 00 02 00 01 E2 00 00 (ts 1765306163, button 2, counter 123874)
 - **Backend:** Uses counter for idempotency/reconciliation.
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1.2.2 0x01 — NFC check-in

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x01
uid	5–8	4-byte card UID
reserved	9–10	0

- **FPort:** 11
 - **When:** Staff NFC check-in (button 0).
 - **Example:** 69 38 6F 33 01 50 B5 94 34 00 00
 - **Backend:** Record check-in.
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1.2.3 0x02 — NFC check-out

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x02
uid	5–8	4-byte card UID
reserved	9–10	0

- **FPort:** 11
 - **When:** Staff NFC check-out. Device updates “Last Cleaned” locally.
 - **Backend:** Record check-out.
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1.2.4 0x03 — NFC registered vote

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x03
button_id	5	0..5
card_data	6–9	4-byte card UID
reserved	10	0

- **FPort:** 11
 - **When:** Staff taps NFC then selects button (registered vote).
 - **Backend:** Persist vote with card UID.
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1.2.5 0x10 — Battery status (heartbeat)

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x10
battery_mv	5–6	uint16 BE (mV)
percent	7	0..100 (0 if unknown)
flags	8	bitfield (0 for now)
reserved	9–10	0

- **FPort:** 20
- **When:** Periodic housekeeping (and on-demand after downlink 0x04).
- **Backend:** Store battery/health; use as heartbeat. Good window to send downlinks (Class A).
- **Note:** v1.2.1 doesn't implement battery percentage, will always be 0.

Heartbeat timing (jitter): Heartbeat runs **once per day** at a device-specific time to spread load. First run of the day (UTC):

`heartbeat_time_UTC = 00:00 UTC + offset_minutes`

`offset_minutes = (DevEUI[7]*256 + DevEUI[6]) % 1440`

DevEUI is 8 bytes (index 0..7); bytes 6 and 7 are the last two. Result is 0..1439 minutes after midnight UTC. Backend can derive each device's daily heartbeat time from DevEUI for scheduling or "last seen" alerts.

1.2.6 0x12 — Counter sync

Field	Bytes	Format
ts	0–3	uint32 BE
evt	4	0x12
button_id	5	0..5
counter	6–8	24-bit BE
reserved	9–10	0

- **FPort:** 20
 - **When:** After rejoin and during housekeeping; one uplink per button (rate-limited).
 - **Backend:** Reconcile per-device per-button counters (e.g. after outage).
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1.2.7 0x11 — Low battery Alarm (reserved)

- Defined in firmware; not yet sent. Same FPort as housekeeping when implemented.
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1.2.8 0xFF — Future

- Reserved for new event types.
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1.3 Downlink: Commands

First byte = command. Device ignores FPort for routing.

Cmd	Code	Payload	Effect
EPD update	0x01	5 bytes: 01 E3 E2 E1 E0 (epoch uint32 BE)	Set “Last Cleaned” on EPD.
EPD refresh	0x02	1 byte: 02	Full e-paper refresh (clear ghosting).
Timezone offset	0x03	3 bytes: 03 HH LL (int16 BE, minutes from UTC)	EEPROM timezone for EPD “Last Cleaned” display.
Status request	0x04	1 byte: 04	Triggers heartbeat (battery + counter-sync); use RX after next uplink for downlink.
Reset counters	0x05	1 byte: 05	Set all button counters to 0.
Factory reset	0x06	1 byte: 06	Clear counters, devnonce, has_joined_once; device reboots (first-boot behavior).

Timezone examples (0x03 payload = 03 + int16 BE minutes from UTC):

Timezone	Offset (min)	Hex payload
New York (EST, winter)	−300	03 FE D4
New York (EDT, summer)	−240	03 FF 10
Denver / Mountain (MST)	−420	03 FE 5C
Denver / Mountain (MDT)	−360	03 FE 98
San Francisco / Pacific (PST)	−480	03 FE 20
San Francisco / Pacific (PDT)	−420	03 FE 5C

To switch a device to another timezone: send one downlink with the 3-byte payload above. Device stores it and uses it when rendering “Last Cleaned” on the EPD (UTC epoch + offset = local time). No reboot required.

How to send: Queue downlink with payload = command byte (+ optional bytes as above). Any FPort (e.g. 10). Unconfirmed typical; device is Class A so downlink only in RX1/RX2 after an uplink.

EPD update example: Epoch 1771131600 → 01 69 91 51 90. <https://www.epochconverter.com/hex>

1.4 Notes

- Downlink only in RX1/RX2 after device uplink. For on-demand status, send **0x04**; device will send battery + counter-sync; and then backend can reply in that RX window (e.g. 0x01 “Last Cleaned”).
 - Button IDs: 0..5 (6 buttons). Counter: 24-bit, per-button, big-endian in payload.
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1.5 Example payloads (real uplinks)

Decoded samples from device uplinks. All timestamps are UTC epoch (bytes 0–3, big-endian); byte 4 = event code.

FPort	Base64	Hex	Explanation
10	aZ3guwAAAAEAAA=	69 9d e0 bb 00 00 00 00 04 00 00	Button (0x00): ts = 0x699DE0BB (~2026-02-24 11:39 UTC), button_id = 0, counter = 4. Public vote on button 0.
11	aZ3g7wGqqr7AAA=	69 9d e0 ef 01 aa aa bb bb 00 00	NFC check-in (0x01): ts = 0x699DE0EF, uid = AA AA BB BB. Staff check-in.
11	aZ3hCAKqqr7AAA=	69 9d e1 08 02 aa aa bb bb 00 00	NFC check-out (0x02): ts = 0x699DE108, uid = AA AA BB BB. Staff check-out.
11	aZ3hKQMCqqq7uwA=	69 9d e1 29 03 02 aa aa bb bb 00	NFC vote (0x03): ts = 0x699DE129, button_id = 2, card = AA AA BB BB. Registered NFC vote for button 2.
20	aZ3YuRIAAAABAAA=	69 9d d8 b9 12 00 00 00 01 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 0, counter = 1 (24-bit BE). Rejoin/heartbeat sync for button 0.
20	aZ3YuRIBAAABAAA=	69 9d d8 b9 12 01 00 00 01 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 1, counter = 1.
20	aZ3YuRICAAABAAA=	69 9d d8 b9 12 02 00 00 01 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 2, counter = 1.
20	aZ3YuRIDAAAAAAA=	69 9d d8 b9 12 03 00 00 00 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 3, counter = 0.
20	aZ3YuRIEAAAAAAA=	69 9d d8 b9 12 04 00 00 00 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 4, counter = 0.
20	aZ3YuRIFAAAFAAA=	69 9d d8 b9 12 05 00 00 05 00 00	Counter sync (0x12): ts = 0x699DD8B9, button_id = 5, counter = 5.
20	aZ3ZwBAM1wAAAAA=	69 9d d9 c0 10 0c d7 00 00 00 00	Battery (0x10): ts = 0x699DD9C0, battery_mv = 0x0CD7 (3287 mV), percent = 0, flags = 0. Heartbeat.

Decoding reference: [0–3] = timestamp (uint32 BE); [4] = event (0x00=button, 0x01/0x02/0x03=NFC in/out/vote, 0x10=battery, 0x12=counter sync); [5..] = event-specific (button_id, counter 24-bit BE, uid, battery_mv uint16 BE, etc.).